Abstract

BounceCoach is an edge-based AR dribble training game that provides real-time spatial feedback and LLM-generated coaching reports.

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Index Terms:
AI coach, Fun drills, AR basketball.

BounceCoach: Real-Time Basketball Dribble Training with On-Site Large Language Model Guidance

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Short Introduction

BounceCoach turns solo dribbling into an AR rhythm game with <40ms edge AI feedback and post-session LLM coaching.

PROBLEM

- Solo practice lacks instant correction
- Coaching is costly & not scalable
- Smart devices miss spatial control + rhythm

SOLUTION

Edge-based AR loop for dribble training:

- Ball tracking + trajectory feedback
- Targets + scoring + effects
- LLM report for personalized coaching

CONTRIBUTIONS

- AR dribble training system for rhythm & spatial control
- Low-cost alternative to motion capture labs
- AI coach that converts metrics into readable guidance

System Overview

Input
Camera + Edge device

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TRACK
Real-time ball detection & trajectory

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PLAY
Dynamic targets + scoring

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FEEDBACK
Rhythm & hand-balance visualization

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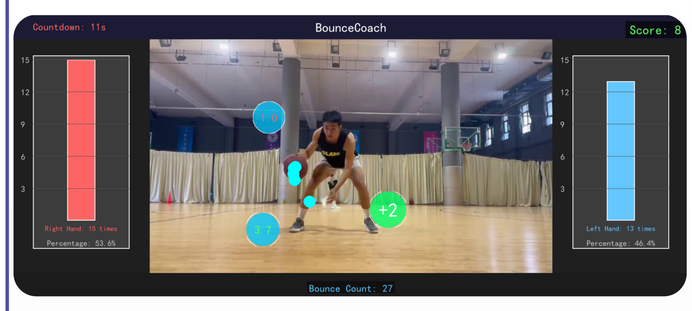
COACH
Personalized post-session report

Key Findings

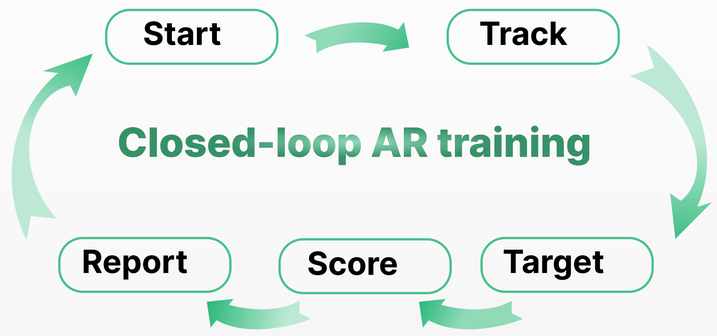
- **<40 ms** latency
- N=24 pilot
- **+28%** consistency
- **1.6×** more fun
- Edge AI (On-device)
- AR game loop

Interaction Design

- Choose hand & duration
- Dynamic targets within reachable space
- Score by hitting targets in time window
- Real-time stats: left/right frequency, rhythm balance



Making basketball training measurable, playable, and accessible.



USER STUDY

- Participants: N= 24
- Conditions: Baseline practice vs. BounceCoach
- Task: short dribbling sessions with target-hitting gameplay
- Measures: dribble consistency + self-reported fun

Participants → Practice → Metrics

Limitations

- Tracking affected by occlusion
- Short-term pilot evaluation
- Limited device testing

NEXT STEP

- Richer drills
- Long-term training study
- Multi-device deployment

RESULTS

BounceCoach improves both performance and engagement in casual dribble practice. In a pilot study, participants achieved +28% higher dribble speed consistency and reported 1.6× more fun compared to baseline training.

